

Electrodynamics

FY549

Lecture Notes for 3rd semester physics and physics and
engineering students

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*Based on D.J. Griffiths, Introduction to Electrodynamics
and D. Tong, Lectures on Electromagnetism*

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How to read these notes:

- Sections in **blue boxes** contain extra material that is helpful or interesting but not strictly necessary.
- Sections in **purple boxes** are for the interested reader who wants to go deeper.
- Sections in **red boxes** are in-class exercises.
- Key equations are marked with **orange boxes**.
- Text marked in **teal** refers to slides shown in the lecture.
- You have plenty of space in the left margin for your own comments.

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1 Maxwell's Equations

Literature: Griffiths Chapter 7.3

Maxwell's equations describe how electric and magnetic fields are generated from charges and currents. Historically, they are the culmination of theory to describe experimental results obtained by Faraday, Ampère, and many others. They form the axioms of classical electrodynamics.

Once we have written them down, we could stop right there — they encode (almost) everything in electrodynamics. We will study their consequences throughout this course, with emphasis on theory. Let us first review them and some of the vector calculus we need to understand them.

We write them in differential form and derive the integral form that you should be familiar with from your electrostatics course.

Consider the electric field $\mathbf{E} = (E_x, E_y, E_z)^T$ and the magnetic field $\mathbf{B} = (B_x, B_y, B_z)^T$.

1.1 Gauss's Law

$$\nabla \cdot \mathbf{E}(\mathbf{x}, t) = \frac{\rho(\mathbf{x}, t)}{\varepsilon_0} \quad (1)$$

Here, the left-hand side is the *divergence* of $\mathbf{E}$, with the nabla operator

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \quad (2)$$

so that

$$\nabla \cdot \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \sum_i \partial_i E_i \equiv \partial_i E_i \quad (3)$$

where in the last step we have used the *Einstein summation convention*: repeated indices are summed over.

Note on index notation

We will frequently use index notation where $\partial_i \equiv \frac{\partial}{\partial x_i}$ with $x_1 = x$, $x_2 = y$, $x_3 = z$. Repeated indices (one upper, one lower, or two of the same in this context) are implicitly summed over: $\partial_i E_i \equiv \sum_{i=1}^3 \partial_i E_i$.

In the second part of the course, it will also become important if we write the indices as subscripts or as superscripts, but this will not bother us now.

Note that $\mathbf{E}$ is a *vector field*: its strength and direction may vary with position and time. The divergence measures whether $\mathbf{E}$ has sources or sinks at a given point.

An example of a vector field with divergence is the electric field of a point charge, see the [slides for an example](#). Another example is the velocity field of a fluid flowing out of a faucet. We will explore this further in the exercises.

Gauss's law states: the source of divergence of $\mathbf{E}$ is the charge density:

- ρ : charge density (charge per unit volume)
- ϵ_0 : vacuum permittivity (ability of $\mathbf{E}$ to permeate vacuum)

$$\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2 \text{N}^{-1} \text{m}^{-2} \quad \left(\epsilon_0 = \frac{10^7}{4\pi c^2} \right) \quad (4)$$

1.1.1 Physical Interpretation: Integral Form

Integrate both sides of Eq. (1) over a volume V :

$$\underbrace{\int_V (\nabla \cdot \mathbf{E}) dV}_{\text{Gauss' theorem}} = \frac{1}{\epsilon_0} \int_V \rho dV \quad (5)$$

$\searrow$
 $= \oint_S \mathbf{E} \cdot d\mathbf{a}$

The right-hand side gives the total charge Q enclosed in V .

Note on notation for volume element dV

Throughout the course, we will use dV interchangeably with d^3x to denote the volume element (Griffith uses $d\tau$). In Cartesian coordinates, this is $d^3x = dV = dx dy dz$.

Applying the *divergence theorem* (Gauss's theorem) to the left-hand side:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad (6)$$

where S is the closed surface enclosing V and $d\mathbf{a}$ points outward. The left-hand side is the *electric flux* through S .

The larger the charge inside V , the larger the flux of $\mathbf{E}$ through the closed surface S .

Note on flux

Flux is not flow. Flow is the average normal component of the field times the surface area.

Importantly, the *shape* of S does not matter — only the total charge enclosed in S does, see Figure 1.

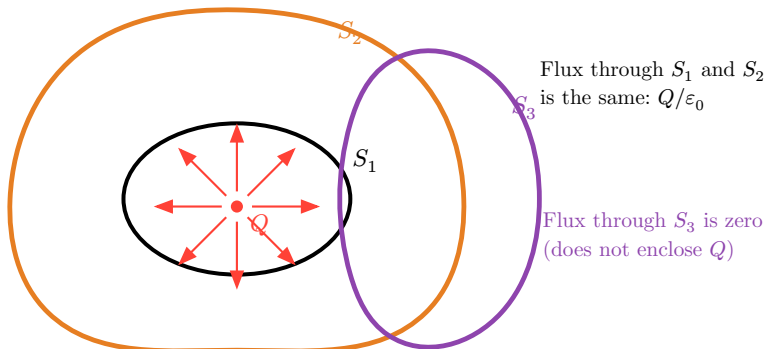


Figure 1: The flux through S_1 and S_2 is the same ($= Q/\epsilon_0$), since both enclose Q . The flux through S_3 is zero: as much flux enters V_3 as leaves it, since S_3 does not enclose the charge.

1.2 Gauss's Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0 \quad (7)$$

This is analogous to Gauss's law in Eq. (1), but for the $\mathbf{B}$ field. It states that there are no magnetic charges (no magnetic monopoles — at least, we have not found them yet).

The net magnetic flux through any closed surface is zero: any outward flux is balanced by inward flux. If you like to visualize the fields with field lines: they are always *closed* for magnetic fields.

This is also true in the static case (magnetostatics).

1.3 Faraday's Law of Induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (8)$$

This is Faraday's law of induction. Note that this is a *vector* equation (in contrast to Gauss's law), so both left- and right-hand sides have three components.

The curl is computed as

$$\nabla \times \mathbf{E} = \det \begin{pmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{pmatrix} = \begin{pmatrix} \partial_y E_z - \partial_z E_y \\ -\partial_x E_z + \partial_z E_x \\ \partial_x E_y - \partial_y E_x \end{pmatrix} \quad (9)$$

The Levi-Civita tensor

In index notation, using the Levi-Civita tensor:

$$(\nabla \times \mathbf{E})_i = \varepsilon_{ijk} \partial_j E_k \quad (10)$$

The Levi-Civita tensor ε_{ijk} is defined as

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{even permutation of } 1, 2, 3 \\ -1 & \text{odd permutation of } 1, 2, 3 \\ 0 & \text{for repeated indices} \end{cases} \quad (11)$$

An *even* permutation is one obtained by cyclic rotation: $123 \rightarrow 231 \rightarrow 312$. You can think of a cyclic rotation as a rotation of a triangle with the numbers 1, 2, 3 at the corners. An *odd* permutation is obtained by swapping two indices: $123 \rightarrow 213 \rightarrow 321 \rightarrow 132 \rightarrow 231 \rightarrow 312$.

1.3.1 Physical Interpretation: Integral Form

Integrate both sides over a surface S :

$$\underbrace{\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a}}_{\text{Stokes' theorem}} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} \stackrel{\text{Holding } S \text{ constant in time}}{=} - \frac{d}{dt} \underbrace{\int_S \mathbf{B} \cdot d\mathbf{a}}_{\text{Magnetic flux } \Phi_B \text{ through surface } S} \quad (12)$$

$$\stackrel{\text{Stokes' theorem}}{=} \oint_C \mathbf{E} \cdot d\mathbf{l}$$

where C is the closed curve bounding S , and $\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{a}$ is the magnetic flux through the surface. The last equality only holds when S is fixed in time, i.e. the surface does not move or deform as time progresses. In that case, the time derivative can be taken outside of the integral.

Collecting results gives the integral form of Faraday's law:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt} \quad (13)$$

Geometrical interpretation of Stokes' theorem

The curl measures the “twist” of a vector field (a whirlpool is a region with high curl). The integral of the flux of the curl through a surface represents the total “amount of swirl”. Stokes' theorem says this equals the circulation — the flow along the boundary. This is illustrated in Figure 2. Crucially, the integral depends only on the boundary C , not on the surface S itself.

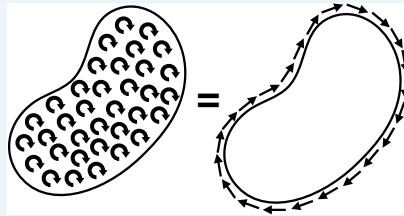


Figure 2: Small circulation loops covering the surface cancel along shared interior edges, leaving only the circulation along the outer boundary C .

The physical content: a time-varying magnetic flux through a surface induces an electric field along the closed path around the surface.

If you place a wire along this path, current will flow as $\mathbf{E}$ exerts force on the charge carriers inside the wire. This is the *electromotive force* (emf):

$$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt} \quad (14)$$

where $\mathbf{f}$ is the force per unit charge responsible for pushing the charge around the circuit.

Lenz's law

The minus sign in Faraday's law encodes Lenz's law: the emf opposes the change in magnetic flux.

This was Faraday's experimental breakthrough: three experiments involving a wire and a magnetic field all yielded a current through the wire, [see the slides for the experimental setups](#). In experiment (a), the wire moves — this is a case of *motional emf*, where the relevant force is the Lorentz force $\mathbf{f} = \mathbf{v} \times \mathbf{B}$. Experiment (b) gives the same result — only relative motion matters. In experiment (c), the wire is not moving ($\mathbf{v} = 0$), so who does the work? The answer: the electric field that is *induced* by the changing magnetic field. This was Faraday's great insight.

1.4 Ampère's Law with Maxwell's Addition

$$\underbrace{\nabla \times \mathbf{B} = \mu_0 \mathbf{J}}_{\text{Ampère's law}} + \underbrace{\mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}}_{\text{Maxwell's addition}} \quad (15)$$

The first term is *Ampère's law*, and the second is *Maxwell's addition* (the displacement current term).

Here:

- $\mathbf{J}$: current density — the amount of current passing through a surface per unit time. The number of charges passing per second through an area element $d\mathbf{S}$ is $\mathbf{J} \cdot \hat{\mathbf{n}} dS$. Units: $[\mathbf{J}] = \text{A} \cdot \text{m}^{-2}$.
- μ_0 : permeability of free space, $\mu_0 \approx 4\pi \times 10^{-7} \text{ N A}^{-2}$.

Definition of μ_0

μ_0 is defined through the force of attraction between two infinite wires with infinitesimal cross section, 1 m apart, carrying 1 A current in the same direction. In natural units: $\mu_0/(4\pi) = \alpha/(2\pi) \cdot \hbar/(e^2 c)$. Here, α is the fine-structure constant, $\hbar$ is the reduced Planck constant (see your quantum mechanics class!), e is the electric charge and c is the speed of light.

1.4.1 Physical Interpretation: Integral Form (Static Case)

In the static case ($\frac{\partial \mathbf{E}}{\partial t} = 0$), integrate over a surface S and apply Stokes' theorem:

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{a} \quad (16)$$

The right-hand side is μ_0 times the total current through S . The direction of C is determined by the right-hand rule with respect to the surface normal $\hat{\mathbf{n}}$.

The magnetic field along C is determined by the total current passing through S .

1.4.2 Why Is Maxwell's Addition Necessary?

The displacement current term ensures the *consistency* of Maxwell's equations.

Take the divergence of Faraday's law Eq. (8):

$$\underbrace{\nabla \cdot (\nabla \times \mathbf{E})}_{=0 \text{ (divergence of curl always zero)}} = -\frac{\partial}{\partial t} \left(\underbrace{\nabla \cdot \mathbf{B}}_{=0 \text{ (2nd Maxwell equation)}} \right) = 0 \quad \checkmark \quad (17)$$

Now take the divergence of Ampère's law in Eq. (15) *without* Maxwell's addition:

$$\underbrace{\nabla \cdot (\nabla \times \mathbf{B})}_{=0} = \mu_0 \nabla \cdot \mathbf{J} \quad (18)$$

This would require $\nabla \cdot \mathbf{J} = 0$ always, which is not true in general (e.g. when charges leave a volume).

With Maxwell's addition:

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E}) = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \rho}{\partial t} \quad (19)$$

which gives the *continuity equation*:

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0 \quad (20)$$

This is an important result that we will use often. It expresses *conservation of charge*: a change in charge density in time is associated with a divergence in the current (charges flowing in or out of the volume).

In integral form:

$$\oint_S \mathbf{J} \cdot d\mathbf{a} = -\frac{dQ}{dt} \quad (21)$$

1.5 Summary: Maxwell's Equations

These are Maxwell's *microscopic* equations: ρ and $\mathbf{J}$ include *all* charges and currents, also “internal” ones like bound charges in atoms and molecules. They are a system of coupled first-order partial differential equations.

In-class exercise

- How many equations do we have (counting components)?
- Why didn't we write them in Cartesian coordinates?

Here they are in one place (combining Eq. (1), Eq. (7), Eq. (8), and Eq. (15)):

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} & (i) \\ \nabla \cdot \mathbf{B} &= 0 & (ii) \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} & (iii) \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} & (iv) \end{aligned} \quad (22)$$

1.5.1 Consequences

Suppose you switch on a current $\mathbf{J}$:

- $\mathbf{J}$ varies in time
- generates a time-variable $\mathbf{B}$ field through Ampère's law (iv)
- generates a time-variable $\mathbf{E}$ field through Faraday's law (iii)
- which in turn modifies $\mathbf{B}$, and so on...

This sounds like an oscillation that propagates — Maxwell's equations are the source of *electromagnetic waves*. We will get to that in the coming weeks.

2 Maxwell's Equations in Matter

Literature: Griffiths Ch. 7.3 (cont'd)

Maxwell's equations Eq. (22) (i)–(iv) are also valid in the presence of matter. However, ρ and $\mathbf{J}$ then include *all* charges and currents, also those that are “bound” in matter — over which we have no direct influence.

It is convenient to rewrite Maxwell's equations in terms of new fields $\mathbf{D}$ and $\mathbf{H}$ that distinguish between *free* charges and currents (ρ_f , $\mathbf{J}_f$) that we can experimentally control, and *bound* charges and currents (ρ_b , $\mathbf{J}_b$).

The bound charges ρ_b and currents $\mathbf{J}_b$ are produced as $\mathbf{E}$ and $\mathbf{B}$ *polarize* and *magnetize* a medium (e.g. electrons and protons in a material get slightly shifted against each other).

At the microscopic level, $\mathbf{E}$ and $\mathbf{B}$ are extremely complex: close to particles, fields become extremely large; further away, closer to another particle, the field might flip direction; at finite temperature, particles undergo thermal fluctuations and fields change continuously.

We focus on the *macroscopic* level — averaged over large enough regions such that we are not affected by microscopic effects. This approach of finding relevant information using an *effective* (averaged) description instead of accounting for all microscopic physics is ubiquitous in physics.

2.1 Dielectrics and Polarization

In-class exercise

What happens inside matter when you apply an electric field?

Answer: it depends on the material.

In a **conductor**, there are lots of free electrons. Any electric field is compensated very fast, so $\mathbf{E} = 0$ inside the conductor. (There will still be a current when you hook up a battery, but inside the wire $\mathbf{E} = 0$. We will look at conductors later.)

In **dielectrics** (non-conducting materials), there are no free charge carriers — electrons are bound in atoms. When you apply an electric field, the positions of electrons (negatively charged) and nuclei (positively charged) shift (in which direction will the particles shift with respect to each other?), leading to small *dipole moments*:

$$\mathbf{p} = q \cdot \mathbf{d} = \alpha \cdot \mathbf{E} \quad (23)$$

where α is the *polarizability* (a scalar or tensor).

Molecules like water that already have a dipole moment will experience a torque and align with the $\mathbf{E}$ field.

The result: lots of dipoles pointing along $\mathbf{E}$ — the material gets *polarized*. A convenient measure is the **polarization**:

$$\mathbf{P} = \text{dipole moment per unit volume}$$

2.1.1 Bound Charges

We can also understand polarization in terms of *bound charges* or the bound charge density ρ_b

When the induced dipoles all align, the interior charges cancel in pairs, as depicted in Figure Figure 3. The net effect is a surface charge — what we call *bound charge*.

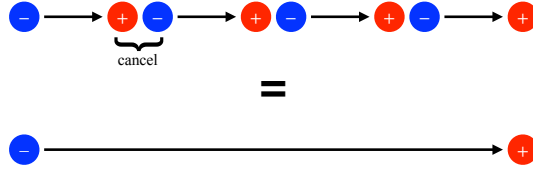


Figure 3: Illustration of aligned dipoles inside a material. The inner charges cancel leaving effectively a bound surface charge.

One can show (Griffiths Ch. 4) that bound charges and polarization are connected through

$$-\nabla \cdot \mathbf{P} = \rho_b \quad (24)$$

The total charge density is

$$\rho = \rho_b + \rho_f \quad (25)$$

where ρ_f is the *free* charge density that we can manipulate in the lab.

Example on Polarization

Imagine a sphere with a uniform distribution of positive and negative charges, as shown in Figure Figure 4.

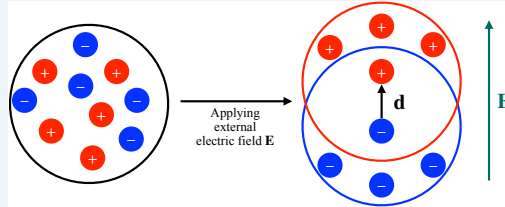


Figure 4: When an electric field is applied, the positive charges move in the direction of $\mathbf{E}$ while the negative charges move in the opposite direction. As a result, the spheres are shifted against each other.

The electric field in the overlapping region is given by

$$\mathbf{E}_{\text{overlap}} = -\frac{1}{4\pi\epsilon_0} \frac{q \cdot \mathbf{d}}{R^3}, \quad (26)$$

where R is the radius of the sphere (believe me or check in Griffiths Ch. 2 and 4).

The electric field can be expressed in terms of the polarization $\mathbf{P}$ replacing the dipole moment $\mathbf{p}$ (with charge q the net positive charge) with the polarization $\mathbf{P}$:

$$\mathbf{p} = q \cdot \mathbf{d} = \mathbf{P} \cdot V = \mathbf{P} \cdot \left(\frac{4}{3}\pi R^3 \right), \quad (27)$$

which is uniform accross the sphere. Plugging in:

$$\Rightarrow \mathbf{E}_{\{\text{overlap}\}} = -\frac{1}{3\epsilon_0} \mathbf{P}. \quad (28)$$

Outside the sphere, for distances $|\mathbf{r}| \gg |\mathbf{d}|$ we have a dipole with potential

$$U = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{|\mathbf{r}|^3} \quad (29)$$

The total electric field from the uniform polarization is shown in Figure 5.

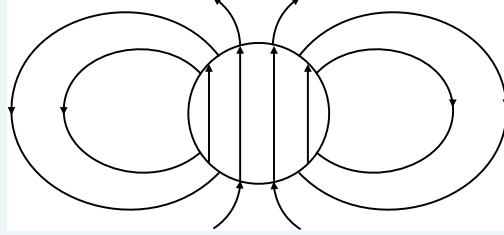


Figure 5: The total electric field from the uniform polarization. The field is uniform inside the sphere and behaves like a dipole outside.

2.1.2 The Electric Displacement $\mathbf{D}$

We define the *electric displacement* $\mathbf{D}$ whose divergence equals the free charge density:

$$\nabla \cdot \mathbf{D} = \rho_f \quad (30)$$

This is like Gauss's law but only for free charges and without ϵ_0 . Starting from the divergence of the total $\mathbf{E}$ field:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{\epsilon_0} (\rho_f + \rho_b) \\ &= \frac{1}{\epsilon_0} \nabla \cdot \mathbf{D} - \frac{1}{\epsilon_0} \nabla \cdot \mathbf{P} \\ &= \frac{1}{\epsilon_0} \nabla \cdot (\mathbf{D} - \mathbf{P}) \end{aligned} \quad (31)$$

This gives us the definition of $\mathbf{D}$:

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad (32)$$

This is actually the definition of $\mathbf{D}$, motivated here from consistency with Maxwell's equations.

Next lecture: we will see what happens in matter when you apply a magnetic field, and rewrite Maxwell's equations fully in terms of free charges and currents.

2.2 Magnetization

Last lecture we looked at the electric field in dielectrics and introduced the *electric displacement*

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad \nabla \cdot \mathbf{D} = \rho_f. \quad (33)$$

We saw that dielectrics become polarized when an external field is applied (note that $\mathbf{E}$ above is the *total* field).

We now ask the analogous question for magnetic fields.

In-class exercise

What happens when you place a piece of material in a magnetic field?

Again, it depends on the type of material.

Some materials, like iron, become magnets themselves and can even stay magnetized once the external field is switched off. Such substances (iron, nickel, cobalt) are called **ferromagnets** — we will not discuss them further in this course.

But other materials are also affected by external fields, and are divided into:

- **Paramagnets:** acquire a magnetization *parallel* to $\mathbf{B}$.
- **Diamagnets:** acquire a magnetization *antiparallel* to $\mathbf{B}$, though the magnetization is much smaller than in ferromagnets (you can't pick up wood with a magnet).

2.2.1 Microscopic Origin: Magnetic Dipoles

Let's look at paramagnets and diamagnets more closely. In fact, all magnetic phenomena are caused by tiny currents:

- electrons that have *spin* (imagine them as spinning charged spheres, even though this picture is not correct in a quantum-mechanical sense),
- electrons that *orbit* around the nucleus.

Both phenomena generate current loops that we can treat as magnetic dipoles, sketched in Figure Figure 6. The dipole moment is given by

$$\mathbf{m} = I \mathbf{A} \quad (34)$$

where I is the current circulating around the loop of area A , and $\mathbf{A} = A\hat{\mathbf{n}}$ points along the loop's normal (direction set by the right-hand rule).

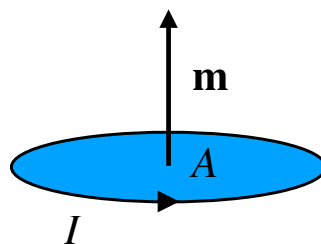


Figure 6: A magnetic dipole moment $\mathbf{m}$ is generated by a current loop. The direction of $\mathbf{m}$ is given by the right-hand rule.

2.2.2 Paramagnetism

Without an external field, these dipoles point in random directions. Once you apply a field, the dipoles experience a torque that will line them up parallel to the field — this torque accounts for paramagnetism.

You might expect this to be a universal phenomenon, since every spinning electron constitutes a dipole.

However, in quantum mechanics we have the **Pauli exclusion principle**, which states that two electrons (or more generally, *fermions* — particles with half-integer spin) cannot occupy the same quantum state. This tends to lock electrons with opposite spin together in atoms, so that their torques neutralize.

As a result, paramagnetism occurs mostly in atoms or molecules with an *odd* number of electrons. But even in such materials, random thermal fluctuations compete with the ordering of the dipoles.

2.2.3 Diamagnetism

Orbits of electrons also constitute tiny current loops, but these are randomly oriented. In the presence of a field, an orbiting electron experiences an additional Lorentz force that will either speed it up or slow it down. This changes its dipole moment *antiparallel* to the field — this is diamagnetism.

Diamagnetism is typically weaker than paramagnetism, and mainly observed in atoms/molecules with an *even* number of electrons (where paramagnetism is largely cancelled by the Pauli exclusion principle).

2.2.4 The Magnetization M

Similarly to polarization, we can thus define the **magnetization**:

$$M = \text{magnetic dipoles per unit volume}$$

The magnetization is connected to a **bound current density** J_b (produced by the electron spins and orbits) through

$$J_b = \nabla \times M \quad (35)$$

[See the slides for an illustration.](#)

2.2.5 The Auxiliary Field H

The currents we control directly — e.g. by increasing a voltage — are called **free currents**, and they produce a field H , analogous to the electric displacement D in the electrostatic case:

$$\nabla \times H = J_f. \quad (36)$$

In practice, H is used far more frequently than D . Usually H is defined through

$$H = \frac{1}{\mu_0} B - M \quad (37)$$

Note

Unlike B , the field H is not generally divergence-free: $\nabla \cdot H = -\nabla \cdot M$, which need not vanish.

2.2.6 The Polarization Current

So far, the relation $J_b = \nabla \times M$ was only established for the electro-/magnetostatic case (only in this case we have $J = J_f + J_b$). In the time-dependent case, we get an

additional contribution to the current density: a change in polarization also drives a **polarization current** $\mathbf{J}_p$:

$$\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t} \quad (38)$$

What's the physical meaning of this current? It has nothing to do with the bound current $\mathbf{J}_b$; $\mathbf{J}_b$ is associated with the magnetization of the material which involves the spin and orbital motion of electrons. In contrast, the polarization current $\mathbf{J}_p$ is the result of the linear motion of charge as the polarization changes. For example: if $\mathbf{P}$ points in the $+x$ direction and increases, positive charges will move slightly in the $+x$ direction while negative charges move slightly in the $-x$ direction. The cumulative effect of these tiny shifts is the current $\mathbf{J}_p$.

This current is essential to ensure conservation of bound charge. We can check this against the continuity equation (Eq. (20)):

$$\nabla \cdot \mathbf{J}_p = \nabla \cdot \frac{\partial \mathbf{P}}{\partial t} = \frac{\partial}{\partial t} (\nabla \cdot \mathbf{P}) = -\frac{\partial \rho_b}{\partial t}, \quad (39)$$

where we used the definition of the polarization current (Eq. (38)). In the last step we used the relation between polarization and bound charge, Eq. (24). This is exactly the continuity equation for the bound charge density ρ_b , as required.

Does a changing magnetization lead to a charge accumulation?

No — a changing magnetization changes the bound current, but

$$\nabla \cdot \mathbf{J}_b = \underbrace{\nabla \cdot (\nabla \times \mathbf{M})}_{=0 \text{ (divergence of curl always zero)}}, \quad (40)$$

so a changing $\mathbf{M}$ redistributes current but never accumulates net charge anywhere. (Note that this is not to be confused with the polarization current $\mathbf{J}_p$ above, which is associated with the polarization $\mathbf{P}$, not the magnetization.)

2.2.7 Ampère's Law in Matter

The total current density now has three contributions:

$$\begin{aligned} \mathbf{J} &= \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p \\ &= \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \end{aligned} \quad (41)$$

Let's write Ampère's law with Maxwell's addition, Eq. (22) (iii), in terms of $\mathbf{H}$ and $\mathbf{D}$ instead of $\mathbf{B}$ and $\mathbf{J}$. Starting from

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad (42)$$

we substitute $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$ (from Eq. (37)) on the left-hand side, and Eq. (41) on the right-hand side:

$$\underbrace{\mu_0 \nabla \times (\mathbf{H} + \mathbf{M})}_{\text{using definition of } \mathbf{H}} = \underbrace{\mu_0 \left(\mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \right)}_{\text{using definition of } \mathbf{J}} + \underbrace{\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}}_{\text{will combine into } \mathbf{D}} \quad (43)$$

The $\mu_0 \nabla \times \mathbf{M}$ term appears on both sides and cancels. Dividing by μ_0 :

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{P}}{\partial t} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (44)$$

Using the definition of $\mathbf{D}$, Eq. (32), the last two terms combine into $\frac{\partial \mathbf{D}}{\partial t}$:

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \quad (45)$$

Faraday's law Eq. (22) (iii) and Gauss's law for magnetism Eq. (22) (ii) stay unchanged — they don't involve currents or a free/bound charge distinction.

2.3 Macroscopic Maxwell's Equations

We arrive at an alternative formulation of Maxwell's equations, valid inside matter, sometimes called the **macroscopic** Maxwell's equations:

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho_f & \text{(i)} \\ \nabla \cdot \mathbf{B} &= 0 & \text{(ii)} \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} & \text{(iii)} \\ \nabla \times \mathbf{H} &= \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} & \text{(iv)} \end{aligned} \quad (46)$$

plus the constitutive relations

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad \text{and} \quad \mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}. \quad (47)$$

Alternatively, one can stick with the *microscopic* Maxwell's equations (Eq. (22)) in terms of $\mathbf{E}$ and $\mathbf{B}$ only, as long as the sources ρ and $\mathbf{J}$ include *all* charges and currents — free and bound.

Interlude: Connection to research — Axions

Axions could solve two pressing problems in particle physics and cosmology:

- The nature of *dark matter*
- The non-observation of an electric dipole moment of the neutron (the *strong CP problem*)

If axions existed, they could couple to photons in the presence of $\mathbf{B}$ -fields. If axions are dark matter, they would surround us and *modify Maxwell's equations*. Gauss's law would get an additional source term from the spatial gradient of the axion field:

$$\rho \rightarrow \rho - \frac{g_{a\gamma}}{\mu_0 c} \mathbf{B} \cdot \nabla a \quad (48)$$

where $g_{\{a\gamma\}}$ is the axion-photon coupling constant and a the axion field. Ampère's law similarly gets modified:

$$\mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \rightarrow \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} + \frac{g_{a\gamma}}{\mu_0 c} \left[\frac{\partial a}{\partial t} \mathbf{B} - \mathbf{E} \times \nabla a \right] \quad (49)$$

Even without charges and currents, you could generate electromagnetic fields through axion dark matter! Typically, the spatial variations of the axion field are negligible, so the gradient term can be dropped leaving the axion current as the new dominating source term,

$$\mathbf{J}_a = \frac{g_{a\gamma}}{\mu_0 c} \frac{\partial a}{\partial t} \mathbf{B}. \quad (50)$$

The axion field oscillates at a frequency $\omega_a = m_a \frac{c^2}{\hbar}$ set by the axion mass m_a , which is unknown but expected to be very small. This leads to an oscillating axion current in the presence of a static magnetic field, which is the basis for many axion dark matter searches also here at SDU.

2.4 Linear Media and Susceptibility

Often, the displacement and magnetizing fields are written in terms of the **permittivity** ε and **permeability** μ of the material:

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad \mathbf{H} = \frac{1}{\mu} \mathbf{B}. \quad (51)$$

ε and μ can be expressed in terms of the electric and magnetic **susceptibilities** χ_e, χ_m :

$$\varepsilon = \varepsilon_0(1 + \chi_e), \quad \mu = \mu_0(1 + \chi_m). \quad (52)$$

The dielectric constant

The (relative) **dielectric constant** is

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = 1 + \chi_e. \quad (53)$$

χ_e and χ_m are dimensionless numbers, [see the slides for typical values](#).

Linear media have a linear relation between polarization/magnetization and the fields:

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}, \quad \mathbf{M} = \chi_m \mathbf{H}. \quad (54)$$

- For **homogeneous** materials, ε and μ are constants; for **inhomogeneous** materials, they are functions of position.
- For **isotropic** materials, ε and μ are scalars; for **anisotropic** materials (like crystals), they are tensors.

2.5 Boundary Conditions

Starting from Maxwell's equations in matter, we now ask: what happens at the boundary between two media with different permittivities ϵ_1, ϵ_2 and permeabilities μ_1, μ_2 ? For now we only look at the general boundary conditions that the electric and magnetic fields must fulfil.

We will soon see that electromagnetic fields are propagating waves, and that these boundary conditions lead to reflection, transmission, and refraction — e.g. light passing from air into glass or water.

In general, $\mathbf{E}$, $\mathbf{D}$, $\mathbf{B}$, and $\mathbf{H}$ are all discontinuous at the boundary between two media.

2.5.1 Perpendicular Components: $\mathbf{D}$ and $\mathbf{B}$

The form of the discontinuity can be derived from Maxwell's equations in integral form. Starting from Gauss's law, (i) in Eq. (46), and applying the divergence theorem:

$$\int_V (\nabla \cdot \mathbf{D}) dV = \oint_S \mathbf{D} \cdot d\mathbf{a} = Q_{f, \text{enc}}, \quad (55)$$

where $Q_{f, \text{enc}}$ is the total free charge enclosed by the surface S . This holds for *any* closed surface — including a thin “pillbox” straddling the boundary between the two media, see Figure Figure 7.

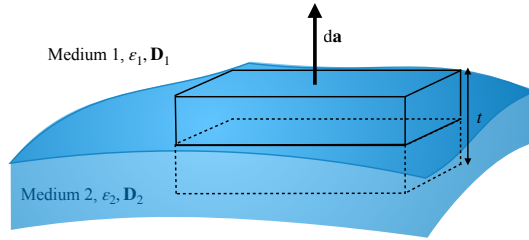


Figure 7: A thin pillbox straddling the boundary between two media. The outward normal vectors $\hat{\mathbf{n}}$ point into medium 1 and medium 2, respectively.

Consider the limit where the pillbox thickness $t \rightarrow 0$: the integral only gets a contribution from the top and bottom faces (each of area a , with outward normal vectors $\pm \hat{\mathbf{n}}$ pointing into medium 1 and medium 2 respectively):

$$\oint_S \mathbf{D} \cdot d\mathbf{a} = (\mathbf{D}_1 \cdot \hat{\mathbf{n}})a - (\mathbf{D}_2 \cdot \hat{\mathbf{n}})a = Q_{f, \text{enc}} = \sigma_f \cdot a, \quad (56)$$

where σ_f is the free surface charge density. Dividing by a :

$$D_1^\perp - D_2^\perp = \sigma_f \quad (57)$$

where $\perp$ denotes the component perpendicular to the boundary. For linear media (Eq. (51)) with no free surface charge:

$$\epsilon_1 \mathbf{E}_1^\perp = \epsilon_2 \mathbf{E}_2^\perp. \quad (58)$$

Similarly, from Gauss's law for magnetism, (ii) in Eq. (46), we have $\oint_S \mathbf{B} \cdot d\mathbf{a} = 0$. The same pillbox argument then gives:

$$B_1^\perp = B_2^\perp \quad (59)$$

The perpendicular component of $\mathbf{B}$ is always continuous.

2.5.2 Parallel Components: $\mathbf{E}$ and $\mathbf{H}$

Now we turn to the two Maxwell equations that involve curls. Using Stokes' theorem, for a closed path C around S , we find the integral forms of (iii) and (iv) in Eq. (46),

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = \oint_C \mathbf{E} \cdot d\mathbf{l}, \quad (60)$$

and similarly

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{a} = \oint_C \mathbf{H} \cdot d\mathbf{l} = I_{f, \text{enc}} + \frac{d}{dt} \int_S \mathbf{D} \cdot d\mathbf{a}, \quad (61)$$

where $I_{f, \text{enc}}$ is the total free current enclosed by the loop.

Consider a thin rectangular Amperian loop with sides of length Δl straddling the boundary, with height h , see Figure 8. In the limit $h \rightarrow 0$, the area S enclosed by the loop vanishes, so the flux term on the right-hand side of Faraday's law vanishes:

$$\mathbf{E}_1 \cdot \Delta \mathbf{l} - \mathbf{E}_2 \cdot \Delta \mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a} \xrightarrow{h \rightarrow 0} 0, \quad (62)$$

so

$$\mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel \quad (63)$$

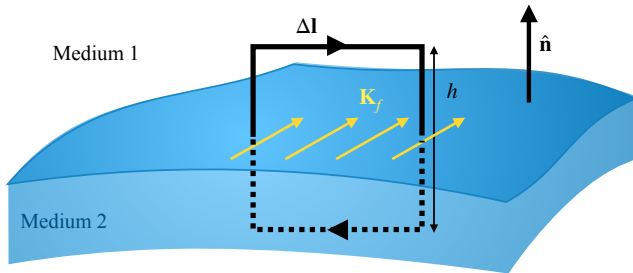


Figure 8: A thin rectangular Amperian loop straddling the boundary between two media.

The parallel component of $\mathbf{E}$ is always continuous across the boundary (note that $\mathbf{E}^\parallel$ is a two-dimensional vector, living in the plane of the boundary).

For Ampère's law (Eq. (61)), the total free current enclosed by the loop can be written in terms of a free surface current density $\mathbf{K}_f$:

$$I_{f, \text{enc}} = \mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \Delta \mathbf{l}), \quad (64)$$

where $\hat{\mathbf{n}} \times \Delta \mathbf{l}$ is the vector perpendicular to the loop, so that $\mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \Delta \mathbf{l})$ picks out the component of $\mathbf{K}_f$ passing through the loop. Using the cyclic property of the scalar triple product, $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$:

$$I_{f, \text{enc}} = \Delta l \cdot (\mathbf{K}_f \times \hat{\mathbf{n}}), \quad (65)$$

so that

$$\mathbf{H}_1 \cdot \Delta \mathbf{l} - \mathbf{H}_2 \cdot \Delta \mathbf{l} = I_{f, \text{enc}} = \underbrace{\Delta \mathbf{l} \cdot \mathbf{K}_f \times \hat{\mathbf{n}}}_{\text{free surface current density}}. \quad (66)$$

Pulling out the common factor $\Delta \mathbf{l}$, we find

$$\mathbf{H}_1^{\parallel} - \mathbf{H}_2^{\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}. \quad (67)$$

The parallel component of $\mathbf{H}$ has a discontinuity proportional to the free surface current density. For linear media (Eq. (51)) with no surface currents:

$$\frac{1}{\mu_1} \mathbf{B}_1^{\parallel} = \frac{1}{\mu_2} \mathbf{B}_2^{\parallel}. \quad (68)$$

Summary: boundary conditions

Collecting the four results derived above (Eq. (57), Eq. (59), Eq. (63), and Eq. (67) respectively), the fields on either side of a boundary between two media are related by:

$$\begin{aligned} D_1^{\perp} - D_2^{\perp} &= \sigma_f & (\text{perpendicular}) \\ B_1^{\perp} &= B_2^{\perp} & (\text{perpendicular}) \\ \mathbf{E}_1^{\parallel} &= \mathbf{E}_2^{\parallel} & (\text{parallel}) \\ \mathbf{H}_1^{\parallel} - \mathbf{H}_2^{\parallel} &= \mathbf{K}_f \times \hat{\mathbf{n}} & (\text{parallel}) \end{aligned} \quad (69)$$

For **linear media** (Eq. (51)) with no free surface charge or current ($\sigma_f = 0$, $\mathbf{K}_f = \mathbf{0}$), the perpendicular and parallel conditions on $\mathbf{D}$ and $\mathbf{H}$ simplify to relations purely between $\mathbf{E}$ - and $\mathbf{B}$ -fields (Eq. (58), Eq. (59), Eq. (63), and Eq. (68) respectively):

$$\begin{aligned} \varepsilon_1 \mathbf{E}_1^{\perp} &= \varepsilon_2 \mathbf{E}_2^{\perp} & (\text{perpendicular}) \\ B_1^{\perp} &= B_2^{\perp} & (\text{perpendicular}) \\ \mathbf{E}_1^{\parallel} &= \mathbf{E}_2^{\parallel} & (\text{parallel}) \\ \frac{1}{\mu_1} \mathbf{B}_1^{\parallel} &= \frac{1}{\mu_2} \mathbf{B}_2^{\parallel} & (\text{parallel}) \end{aligned} \quad (70)$$

Note that $\mathbf{B}^{\perp}$ and $\mathbf{E}^{\parallel}$ are always continuous — they don't depend on the medium at all.

3 Electromagnetic Waves

Literature: Griffiths Ch. 9.1 and 9.2

Last lectures we reviewed Maxwell's equations and derived the boundary conditions at the interface between two media. We will now study how electromagnetic waves arise from Maxwell's equations. Let's first review waves in general.

3.1 Waves: A Review

In-class exercise

What is a wave?

In simplest form: a **disturbance that propagates with a fixed shape at constant velocity**. This neglects, among other things:

- *absorption* (the wave amplitude will diminish),
- *dispersive effects* (different frequencies travel at different speeds).

In one dimension, a wave is a solution to the **wave equation**:

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2} \quad (71)$$

where v is the velocity of the wave.

The solution is any function of the form

$$f(z, t) = g(z - vt) + h(z + vt) \quad (72)$$

In-class exercise

What is the physical interpretation of g and h ?

$g(z - vt)$ is a wave traveling to the *right*: the displacement found at z at time t is found again at $z + v\Delta t$ at the later time $t + \Delta t$, see Figure 9. Likewise, $h(z + vt)$ is a wave traveling to the *left*.

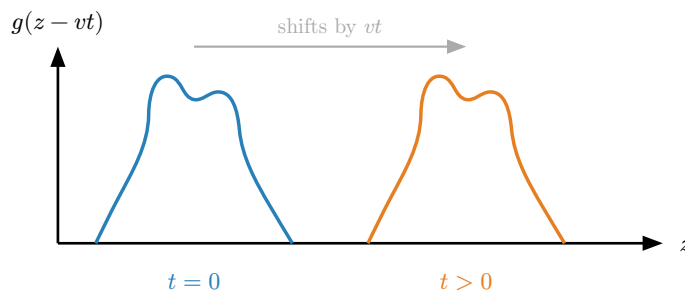


Figure 9: A wave $g(z - vt)$ traveling to the right: the same shape reappears, shifted by vt , at a later time.

3.1.1 Sinusoidal Waves

A particularly important special case is the sinusoidal wave:

$$f(z, t) = A \cos(k(z - vt) + \delta) \quad (73)$$

Amplitude wave number phase

To see that this is a wave: think of it at fixed t at different z , and at fixed z at different times t , then consider what happens if you change both z and t .

Terminology

- **Wavelength** $\lambda = 2\pi/k$: distance between two maxima.
- **Period** $T = 2\pi/(kv)$: in one period, a point goes through a full cycle (e.g. from $+A$ to $-A$ and back to $+A$).
- **Frequency** $\nu = 1/T = (kv)/(2\pi) = v/\lambda$: number of oscillations per unit time.
- **Angular frequency** $\omega = 2\pi\nu = kv$: number of radians per unit time.

Using $\omega = kv$, we can write the sinusoidal wave as

$$f(z, t) = A \cos(kz - \omega t + \delta) \quad (74)$$

We can use Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, to write a wave in complex notation:

$$f(z, t) = \text{Re}(Ae^{i(kz - \omega t + \delta)}) \quad (75)$$

This notation is very useful for solving the wave equation.

The superposition principle

The wave equation Eq. (71) is *linear*, so that the sum of two solutions is also a solution. This is the **superposition principle**, and it is essential for combining waves (e.g. via Fourier analysis) and for the interference phenomena we will encounter later in the course.

3.2 Waves in 3D

In 3D, there are different possibilities for the direction of oscillation. If we take the z -direction as the direction of propagation, we can have:

1. **Longitudinal waves**: oscillation in the z -direction. Example: sound waves, as density fluctuations in a medium, [see Figure 10](#).
2. **Transverse waves**: oscillations in directions orthogonal to z .

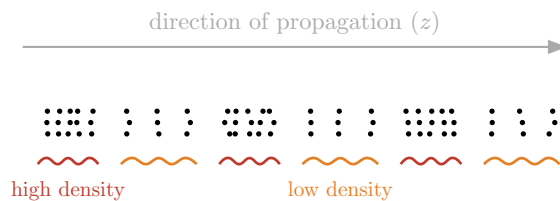


Figure 10: A longitudinal wave: regions of high and low density (compressions and rarefactions) propagate along the direction of the wave.

3.2.1 Polarization

Transverse waves in 3D have **two independent states of polarization**, because there are two directions orthogonal to the propagation direction.

The **polarization vector** $\hat{\mathbf{u}}$ (with $\hat{\mathbf{u}} \cdot \hat{\mathbf{u}} = 1$) denotes the plane of oscillation, and $\hat{\mathbf{u}} \cdot \hat{\mathbf{z}} = 0$ for transverse waves. Figure 11 shows two examples for fixed time t , with $\hat{\mathbf{u}} = \hat{\mathbf{x}}$ and $\hat{\mathbf{u}} = \hat{\mathbf{y}}$ respectively.

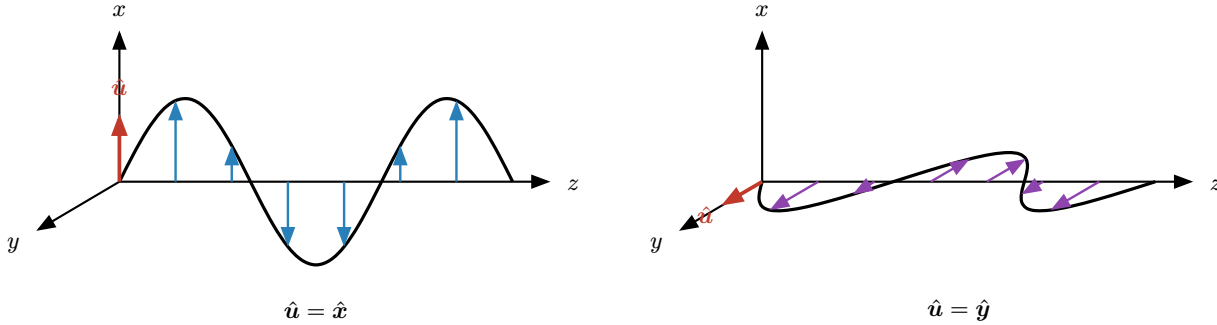


Figure 11: A transverse wave for two linear polarizations: oscillation along $\hat{\mathbf{x}}$ (left) and along $\hat{\mathbf{y}}$ (right). Both are orthogonal to the direction of propagation $\hat{\mathbf{z}}$.

In general, for **linear polarization**:

$$\hat{\mathbf{u}} = \cos \theta \hat{\mathbf{x}} + \sin \theta \hat{\mathbf{y}} \quad (76)$$

with **polarization angle** θ .

Circular polarization

$\hat{\mathbf{u}}$ can also depend on time. If it rotates clockwise around the $\hat{\mathbf{z}}$ axis (as seen looking toward the source), we speak of **right circular polarization**; if it rotates counter-clockwise, of **left circular polarization**.

3.3 Electromagnetic Waves

Recall Maxwell's equations in vacuum, Eq. (22), [see the slides](#). What are the consequences of these equations?

Suppose that you switch on a current $\mathbf{J}(t)$:

- $\mathbf{J}$ will vary in time,
- $\Rightarrow$ generates a $\mathbf{B}$ field, variable in time,
- $\Rightarrow$ generates an $\mathbf{E}$ field, variable in time,
- $\Rightarrow$ will again generate $\mathbf{B}$ through (iv),
- $\Rightarrow$ perpetual interplay between $\mathbf{E}$ and $\mathbf{B}$ through (iii) and (iv).

This already feels like something is oscillating. Let's play a trick to transform Maxwell's equations — first-order differential equations — into *second-order* differential equations, with the hope that they decouple.

We do this by taking the curl of (iii):

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = -\mu_0 \frac{\partial \mathbf{J}}{\partial t} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad (77)$$

where in the last step we exchanged the order of the spatial curl and temporal derivative (coordinates do not depend on time), and then used (iv).

What is the left-hand side? You can show that

The “BAC – CAB” rule for the curl of a curl

Similarly to the vector identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad (78)$$

(“BAC – CAB”), one can show an analogous identity for the differential operator ∇ :

$$\nabla \times (\nabla \times \mathbf{C}) = \nabla(\nabla \cdot \mathbf{C}) - \nabla^2 \mathbf{C}, \quad (79)$$

where ∇^2 is the **Laplacian operator**,

$$\nabla^2 f = \partial_x^2 f + \partial_y^2 f + \partial_z^2 f = \sum_i \partial_i^2 f \equiv \partial_i \partial_i f, \quad (80)$$

acting on each component of $\mathbf{C}$ separately.

Proof (in Einstein notation): look at the i -th component,

$$[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})]_i = \varepsilon_{ijk} A_j (\mathbf{B} \times \mathbf{C})_k = \varepsilon_{ijk} A_j \varepsilon_{klm} B_l C_m. \quad (81)$$

Using the cyclic property $\varepsilon_{ijk} = \varepsilon_{kij}$ and the identity $\varepsilon_{kij} \varepsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$ (which follows from the definition of ε_{ijk} , in the same way as in the note on the Levi-Civita tensor in Chapter 1), this becomes

$$\begin{aligned} \varepsilon_{ijk} A_j \varepsilon_{klm} B_l C_m &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_j B_l C_m \\ &= B_i (A_j C_j) - C_i (A_j B_j) = [\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})]_i. \end{aligned} \quad (82)$$

The same computation with $\mathbf{A} = \mathbf{B} = \nabla$ (so that $A_j \rightarrow \partial_j$) and $\mathbf{C} \rightarrow \mathbf{C}$ gives

$$\begin{aligned} [\nabla \times (\nabla \times \mathbf{C})]_i &= \varepsilon_{ijk} \partial_j \varepsilon_{klm} \partial_l C_m \\ &= \partial_i (\partial_j C_j) - \partial_j \partial_j C_i = [\nabla(\nabla \cdot \mathbf{C}) - \nabla^2 \mathbf{C}]_i. \end{aligned} \quad (83)$$

Using this identity with $\mathbf{C} = \mathbf{E}$, and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$ from (i):

$$\nabla \left(\frac{\rho}{\varepsilon_0} \right) - \nabla^2 \mathbf{E} = -\mu_0 \frac{\partial \mathbf{J}}{\partial t} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (84)$$

Rearranging, with all $\mathbf{E}$ -terms on the left:

$$\underbrace{\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E}}_{\square \mathbf{E}} = -\nabla \frac{\rho}{\varepsilon_0} - \mu_0 \frac{\partial \mathbf{J}}{\partial t} \quad (85)$$

The operator $\square \equiv \mu_0 \varepsilon_0 \frac{\partial^2}{\partial t^2} - \nabla^2$ is the **d’Alembert operator** (or wave operator).

Let’s do the same for Maxwell’s equation (iv). Due to the high symmetry between $\mathbf{E}$ and $\mathbf{B}$ in the vacuum Maxwell equations, we suspect that we will also arrive at a wave equation:

$$\nabla \times (\nabla \times \mathbf{B}) = \nabla \left(\underbrace{\nabla \cdot \mathbf{B}}_{=0} \right) - \nabla^2 \mathbf{B} = \mu_0 \nabla \times \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\underbrace{\nabla \times \mathbf{E}}_{=-\frac{\partial \mathbf{B}}{\partial t}} \right), \quad (86)$$

where we used (ii) and (iv) on the left, and (iv) then (iii) on the right. Rearranging:

$$\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} = \mu_0 \nabla \times \mathbf{J} \quad (87)$$

We have transformed Maxwell's equations into decoupled second-order differential equations for regions with charges and currents. (They are not *always* decoupled — e.g. Ohm's law $\mathbf{J} = \sigma \mathbf{E}$ couples them back together, as is relevant for conductors like radio antennas or the plasma in the sun's corona.)

3.4 Electromagnetic Waves in Vacuum

Let's consider $\mathbf{E}$ and $\mathbf{B}$ fields generated in some region (e.g. by an antenna). How will the fields propagate in vacuum (free space), i.e. away from the region with charges and currents?

There, Eq. (85) and Eq. (87) become:

$$\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} = 0, \quad \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} = 0 \quad (88)$$

These are 3D wave equations! (Eq. (85) and Eq. (87), with sources on the right-hand side, are called *inhomogeneous* wave equations.)

The velocity of the wave is

$$c^2 = \frac{1}{\mu_0 \varepsilon_0} \quad (89)$$

μ_0 and ε_0 can be measured in the lab, and we find $c \approx 2.9979 \times 10^8$ m/s — the speed of light! (Check yourself that $1/\sqrt{\mu_0 \varepsilon_0}$ indeed has units of m/s.)

Unifying electricity, magnetism, and light

We have stumbled onto the nature of light: it is, in fact, a disturbance in the $\mathbf{E}$ and $\mathbf{B}$ fields. With this, Maxwell's equations unify electricity, magnetism, *and* light — one of the great triumphs of 19th century physics.

Note

Maxwell's equations are 8 equations (1 from (i), 1 from (ii), 3 each from (iii) and (iv)), but Eq. (88) only give us 6 (3 components each for $\mathbf{E}$ and $\mathbf{B}$). Not every solution of the wave equation is a solution of Maxwell's equations — it must fulfill some extra constraints, as we will see next.

3.4.1 Plane Wave Solutions

Consider the plane wave ansatz

$$\mathbf{E}(z, t) = \mathbf{E}_0 e^{i(kz - \omega t)}, \quad (90)$$

with constant $\mathbf{E}_0$. In vacuum, $\nabla \cdot \mathbf{E} = 0$, and since $\mathbf{E}(z, t)$ is independent of x, y :

$$\partial_x E_x + \partial_y E_y + \partial_z E_z = 0 \Rightarrow E_{0z}(ik)e^{i(kz-\omega t)} = 0, \quad (91)$$

which only holds if $E_{0z} = 0$. Hence:

EM waves are **transversal**: $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to the direction of propagation.

Take also $\mathbf{B}(z, t) = \mathbf{B}_0 e^{i(kz-\omega t)}$ as a plane wave. From Faraday's law, $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, one finds (check this yourself!) the compact relation

$$\mathbf{B}_0 = \frac{k}{\omega}(\hat{\mathbf{z}} \times \mathbf{E}_0) \quad (92)$$

This tells us:

- $\mathbf{E}$ and $\mathbf{B}$ are both perpendicular to each other (and to $\hat{\mathbf{z}}$), and must both be non-zero.
- $\mathbf{E}$ and $\mathbf{B}$ are *in phase* — they reach their maximum at the same time, since there is no extra phase factor in Eq. (92).
- There are only *electromagnetic* waves (not waves of $\mathbf{E}$ or $\mathbf{B}$ alone).

3.4.2 General Monochromatic Plane Waves

With $\hat{\mathbf{k}}$ the unit wave vector in the direction of propagation, and the dispersion relation $k = \omega/c$, we have for monochromatic plane waves:

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= E_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \hat{\mathbf{u}}, \\ \mathbf{B}(\mathbf{r}, t) &= \frac{1}{c} E_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} (\hat{\mathbf{k}} \times \hat{\mathbf{u}}) = \frac{1}{c} (\hat{\mathbf{k}} \times \mathbf{E}) \end{aligned} \quad (93)$$

with $\hat{\mathbf{u}} \cdot \hat{\mathbf{k}} = 0$.

3.5 Energy and Momentum in EM Waves

The energy per unit volume stored in the fields is

$$u = \frac{1}{2} \left(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) \quad (94)$$

(we will derive this later — for now, check for yourself through dimensional analysis that the units work out).

From Eq. (92) we see that $B^2 = E^2/c^2$, so that

$$u = \varepsilon_0 E^2 = \frac{1}{\mu_0} B^2 \quad (95)$$

— $\mathbf{E}$ and $\mathbf{B}$ contribute equally to u — and

$$u = \varepsilon_0 E_0^2 \cos^2(kz - \omega t + \delta). \quad (96)$$

As the wave propagates, it transports energy. The **energy flux density** (amount of energy transported through a surface per unit time) is given by the Poynting vector:

$$\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}) \quad (97)$$

$\mathbf{S}$ points in the direction of propagation, with $|\mathbf{S}| = cu$.

The momentum density is

$$\mathbf{g} = \frac{1}{c^2}\mathbf{S}, \quad |\mathbf{g}| = \frac{u}{c}. \quad (98)$$

3.5.1 Time-Averaged Quantities

Scale of visible light

Visible light has a wavelength between $\sim 10^2$ and 10^3 nm, which corresponds to a frequency $\nu \sim 10^{15}$ Hz, i.e. a period $T \sim 10^{-15}$ s: very fast oscillations. Usually, we measure for durations much longer than that.

We therefore consider the **time-averaged energy density** over one period $T = 2\pi/\omega$:

In-class exercise

Show that

$$\langle u \rangle = \varepsilon_0 \frac{E_0^2}{T} \int_0^T \cos^2(kz - \omega t + \delta) dt \quad (99)$$

(we will look at time averages like this in the tutorials).

$$\langle u \rangle = \frac{1}{2}\varepsilon_0 E_0^2 \quad (100)$$

since the time average of $\cos^2$ over a full period is $1/2$. This immediately gives the time averages for $\mathbf{S}$ and $\mathbf{g}$ as well.

A historical puzzle

Note that the average energy density is *independent of frequency*. This led to an experimental puzzle — the photoelectric effect — which could not be explained classically, and required quantum mechanics to describe.

3.6 Electromagnetic Waves in Matter

In-class exercise

What are the wave equations for $\mathbf{E}$ and $\mathbf{B}$ in a linear and homogeneous medium, without free currents or charges? Do EM waves propagate slower than in vacuum?

In this case, the macroscopic Maxwell's equations, Eq. (46), read:

$$\begin{aligned} \text{(i)} \quad \nabla \cdot \mathbf{D} &= 0, & \text{(iii)} \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} &= 0, & \text{(iv)} \quad \nabla \times \mathbf{H} &= \frac{\partial \mathbf{D}}{\partial t}. \end{aligned} \quad (101)$$

For **linear media** (Eq. (51)), $\mathbf{D} = \varepsilon \mathbf{E}$ and $\mathbf{H} = \frac{1}{\mu} \mathbf{B}$; for **homogeneous** media, ε and μ are constant (they don't vary from point to point). Plugging in, we find:

$$\begin{aligned} \text{(i)} \quad \nabla \cdot \mathbf{E} &= 0, & \text{(iii)} \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} &= 0, & \text{(iv)} \quad \nabla \times \mathbf{B} &= \mu \varepsilon \frac{\partial \mathbf{E}}{\partial t}. \end{aligned} \quad (102)$$

The only difference to the vacuum case, Eq. (22), is that $\varepsilon_0 \rightarrow \varepsilon$ and $\mu_0 \rightarrow \mu$. Thus, we find EM waves that propagate at speed

$$v = \frac{1}{\sqrt{\varepsilon \mu}} = \frac{c}{n} \quad (103)$$

where

$$n \equiv \sqrt{\frac{\varepsilon \mu}{\varepsilon_0 \mu_0}} \quad (104)$$

is the **index of refraction**.

All previous results carry over to the case of a medium, with $\varepsilon_0 \rightarrow \varepsilon$ and $\mu_0 \rightarrow \mu$. For most materials, $\mu \approx \mu_0$, so that

$$n \approx \sqrt{\varepsilon_r} = \sqrt{\varepsilon/\varepsilon_0}, \quad (105)$$

where ε_r is called the **dielectric constant**, which might depend on frequency. Almost always $\varepsilon_r > 1$, so waves travel *slower* in media.

The energy density and Poynting vector become

$$u = \frac{1}{2} \left(\varepsilon E^2 + \frac{1}{\mu} B^2 \right), \quad \mathbf{S} = \frac{1}{\mu} (\mathbf{E} \times \mathbf{B}). \quad (106)$$

Why does light slow down in matter?

Mathematically, replacing $\varepsilon_0 \rightarrow \varepsilon$ and $\mu_0 \rightarrow \mu$ is a pretty trivial result — but physically, it is quite astonishing. As waves pass through matter, they polarize and magnetize the molecules, producing oscillating dipoles that create their own fields. These new fields combine with the original wave to give a wave at the *same* frequency, but a *different* speed.

This is a non-trivial consequence of linearity, and is the reason materials like glass or water are transparent: the wave that emerges is still a single coherent wave at the original frequency, just slowed down.